An Intelligent Car Insurance Claim Fraud Detection System Using Ensemble Learning

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Abstract*: Insurance fraud has become a serious issue in the automobile insurance sector. Insurance companies lose a lot of money every year because some people submit false or fake claims. Many companies still depend on manual checking systems. These systems take more time, require more manpower, and sometimes fail to identify fraud correctly. Because of these problems, there is a need for an automated and reliable solution. A machine learning-based system is developed to detect fraudulent car insurance claims. The system studies different details such as customer information, vehicle data, and claim-related factors to predict whether a claim is genuine or suspicious. The trained model is connected to a backend API and integrated with a simple user interface for real-time prediction.The application allows users to enter claim information and instantly check whether the claim is likely to be fraudulent. It also maintains claim history and provides an admin dashboard for reviewing and approving claims. Different machine learning algorithms like Decision Tree, Support Vector Machine (SVM), Random Forest, and K-Nearest Neighbors (KNN) were implemented and compared. Their performance was evaluated using accuracy, precision, recall, and F1-score. Among all models, Random Forest showed better performance.The results indicate that the proposed system improves fraud detection and speeds up the claim verification process. This approach can help insurance companies reduce financial losses and make claim processing more efficient.

Keywords: Insurance claims, Car Insurance, Fraud Detection, Machine Learning, Claim Prediction, Artificial Intelligence.

I. INTRODUCTION

The automobile insurance industry is crucial because it helps people and businesses manage financial losses caused by accidents, theft, or vehicle damage. When unexpected incidents happen, insurance gives financial support and reduces stress for vehicle owners. Today, the number of insured vehicles is increasing every year across the world. As a result, insurance companies receive a large number of claims regularly.Along with genuine claims, the number of fraudulent claims has also increased. Some individuals try to misuse insurance policies by submitting false or exaggerated claims to gain money illegally. These fraudulent activities create serious financial losses for insurance companies and also increase their workload. Because of this growing problem, it has become necessary to find better and more efficient ways to identify and prevent insurance fraud [1].

Insurance fraud occurs when policyholders intentionally provide false information, exaggerate damages, or submit fabricated accident reports to obtain unfair compensation. Such fraudulent activities not only impact the profitability of insurance firms but also affect honest customers through increased premium rates and delayed claim processing [2]. Detecting these fraudulent claims at an early stage is essential for maintaining trust and efficiency in the insurance sector.

Traditional fraud detection methods mainly depend on manual investigation and expert verification, which are often time-consuming, expensive, and difficult to scale when dealing with large datasets. Furthermore, modern fraud strategies have become more sophisticated, making it challenging for conventional approaches to identify hidden patterns and suspicious behaviours effectively [4].

To address these challenges, machine learning has emerged as a promising solution for automating fraud detection. By learning patterns from historical insurance claim data, supervised classification algorithms can predict whether a claim is genuine or fraudulent before approval. This reduces investigation workload, improves decision-making accuracy, and accelerates claim settlement processes [6].

Therefore, this paper presents a complete machine learning based framework fraud detection platform that combines machine learning prediction with a MERN stack web application, providing real-time fraud classification, claim management, and administrative decision support.

II. RELATED WORK

Machine learning has become a widely adopted approach in the field of insurance analytics, particularly for fraud detection and claim risk prediction. With the rapid growth of automobile insurance services, insurers are increasingly dependent on intelligent systems that can automatically identify suspicious claims from large volumes of customer and vehicle-related data. Several studies have explored supervised learning techniques to classify insurance claims as fraudulent or genuine based on historical claim patterns [1].

Decision Tree classifiers are commonly used in fraud detection because of their simple rule-based structure and interpretability. These models provide clear decision paths, which help insurance investigators understand why a particular claim is flagged as suspicious. However, Decision Trees may suffer from overfitting when trained on complex or noisy claim datasets [3].

Many researchers have explored different machine learning techniques to detect fraudulent insurance claims. One of the commonly used methods is Support Vector Machine (SVM). SVM is useful because insurance data usually contains many features, and fraud patterns are not always simple. In many cases, fraudulent and genuine claims cannot be separated using a straight line or simple rule. SVM helps by finding better boundaries between classes, especially when the data relationship is complex. However, to get good results, the model must be properly tuned by selecting the right kernel and parameters. [5]

Another popular method in fraud detection is Random Forest. This technique builds multiple decision trees and combines their results to make a final prediction. Because it uses many trees instead of just one, it reduces errors and gives more stable results. Random Forest is also able to handle real-world insurance data, which may include missing information or uneven distribution of fraud cases. Due to these advantages, many studies report that Random Forest performs consistently well in fraud detection tasks [4], [10], [14].

K-Nearest Neighbors (KNN) has also been applied in claim classification. This method works by comparing a new claim with previous claims and checking which ones are most similar. Based on this similarity, it predicts whether the claim is fraudulent or genuine. KNN is easy to understand and implement, but it can become slow when the dataset is very large because it needs to calculate distances with many data points [8].

A common issue mentioned in several studies is the imbalance between fraudulent and genuine claims. Fraud cases are usually much fewer compared to normal claims. Because of this, models may learn more about genuine claims and fail to detect fraud correctly. To improve performance, researchers suggest using techniques such as resampling the dataset or giving more importance to fraud cases during model training.

Even though previous research shows promising results, many studies focus only on testing models in offline environments. In real insurance systems, there is a need for real-time fraud detection. Insurance companies require systems that can instantly predict fraud, maintain claim records, and support administrative decisions. To address this gap, the proposed work develops a complete real-time fraud detection system using Decision Tree, SVM, Random Forest, and KNN models for automobile insurance claim verification.

III. DATASET DESCRIPTION

A. Summary of the Dataset

For this analysis, this study uses the publicly available Car Insurance Claim dataset from Kaggle. It contains structured automobile insurance claim records collected from real-world policyholders. Each record represents an individual insurance claim along with customer demographics, vehicle-related details, accident circumstances, and financial claim information.

The dataset consists of approximately 1,000 claim instances with nearly 39 input features, including both numerical and categorical variables. An 80:20 split was applied, ensuring that the models were trained on the majority of the claim records and tested on unseen samples for reliable evaluation as shown in TABLE I.

TABLE I. Data Partition

Partition	Records	Percentages	Purpose
Training	800	80%	Model learning and development
Testing	200	20%	Model validation and evaluation
Total	1000	100%	Complete claim dataset

Figure 1 represents the count distribution of all features in the car_insurance_claim.csv dataset. It shows that the dataset contains multiple customer, policy, incident, and claim-related attributes with sufficient records, making it suitable for insurance fraud prediction.

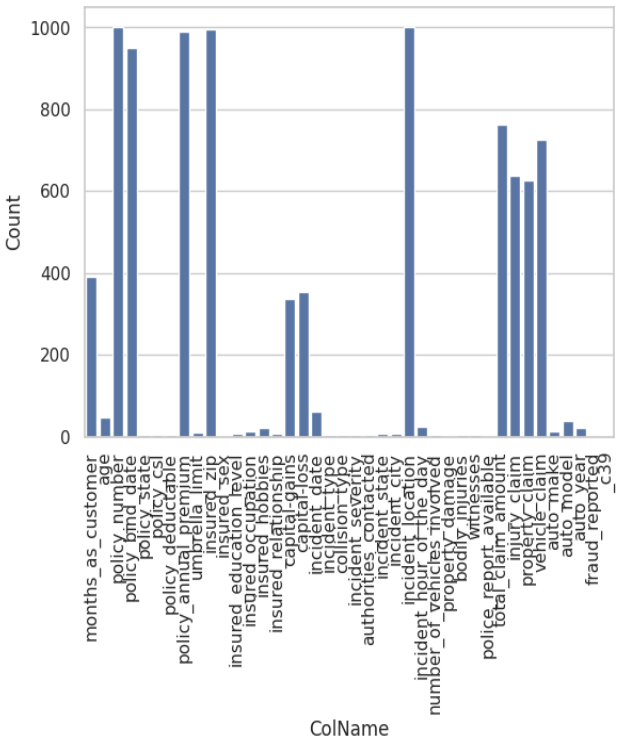

Fig. 1 Feature-wise count distribution of attributes

B. Data Preprocessing

Initially, irrelevant and duplicate columns were removed, and missing values were handled to improve data quality. Categorical features such as policy state, incident type, and vehicle make were converted into numerical form using encoding techniques. Numerical attributes including claim amounts and premium values were scaled to maintain uniform feature ranges, which is important for algorithms such as SVM and KNN.

IV. PROPOSED METHODOLOGY

A. Proposed System Architecture

The proposed Car Insurance Claim Fraud Detection and Prediction System is developed as an end-to-end intelligent platform that integrates machine learning with a MERN-based web application. The system supports real-time claim submission, automated fraud risk prediction, claim tracking, and administrative approval through an interactive dashboard.

The user fills in important details such as personal information, vehicle data, policy details, accident history, and the claim amount using an easy step-by-step form. After submitting the form, the system sends the information to the trained model to check whether the claim seems genuine or not. The prediction is shown instantly and also saved for later review.

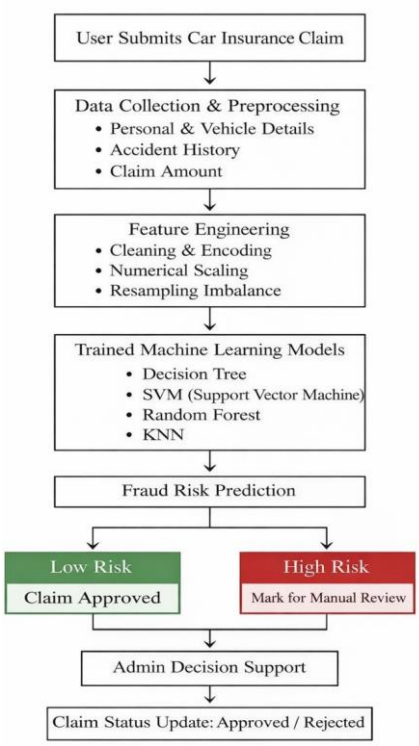

Fig. 2 Proposed System Architecture for Car Insurance Claim Prediction

B. Claim Data Collection and User Interface

The process starts when a user creates an account and logs into the InsurePredict portal. After logging in, the user is taken through a step-by-step claim form that is easy to follow. They need to enter details such as their personal information, vehicle details, insurance policy information, accident history, and the claim amount. By collecting all these details in an organized way, the system gets clear and complete information. This helps improve the accuracy of the fraud prediction and makes the overall process more reliable.

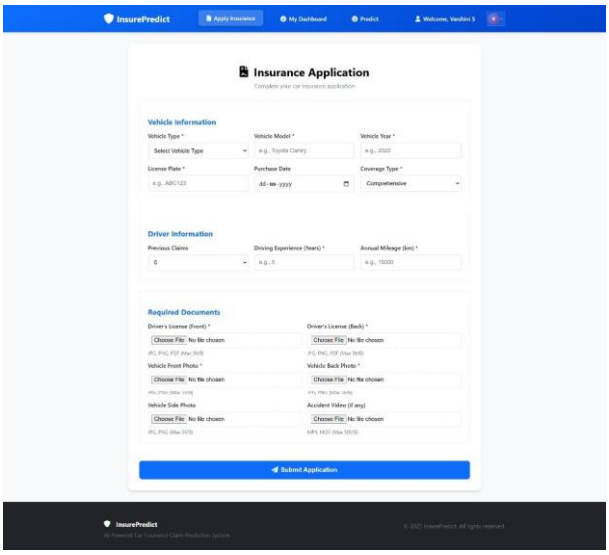

Fig. 3 Insurance Application Form Interface

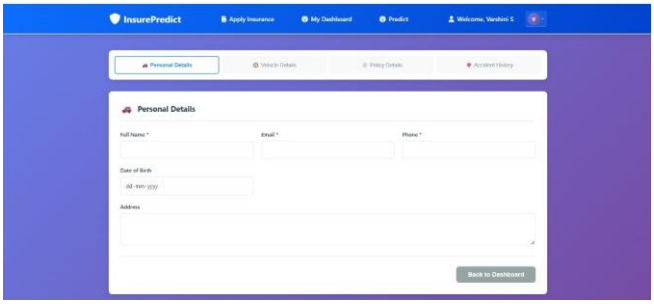

Fig. 4 Personal Details Submission Interface

C. Machine Learning Fraud Prediction Engine

The system mainly works using supervised machine learning methods like Decision Tree, SVM, Random Forest, and KNN. These models are trained using previous insurance claim records so they can understand how fraudulent claims usually look. Once a new claim is submitted, the system checks the details and predicts whether it is genuine or suspicious. Among all the models tested, Random Forest gave more consistent and accurate results because it combines multiple decision trees and handles complex patterns better.

D. Admin Dashboard and Decision Support

The system includes an Admin Dashboard that helps insurance officers easily keep track of all claim applications. It shows important details like how many claims are approved, rejected, or still pending. Officers can check the system's prediction results, look into suspicious cases if needed, and make final decisions smoothly. This makes the claim handling process more organized and faster.

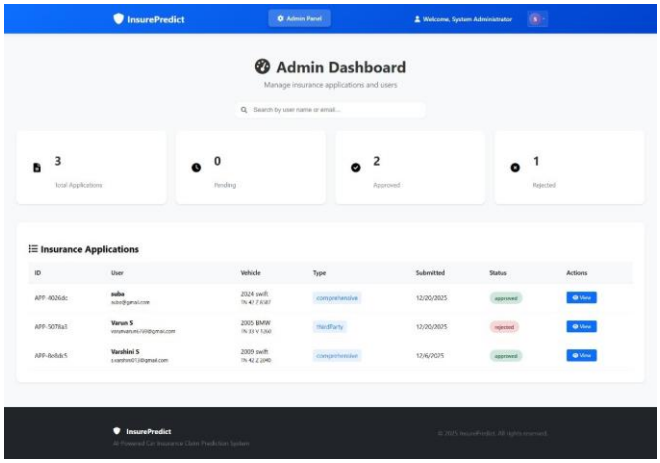

Fig. 5 Admin Dashboard for monitoring and managing claims.

E. Claim Logging and Reporting

Whenever a claim is submitted, the system saves all the details along with the fraud prediction result and any action taken by the admin. Each record is stored with the date and time so that everything can be tracked properly later. This helps maintain clear records and makes the process more transparent. It also allows the company to review past claims, generate reports, and understand fraud patterns over time.

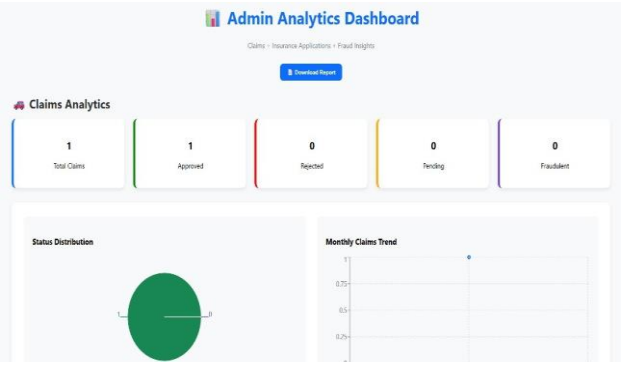

Fig. 6 Admin Analytics Dashboard

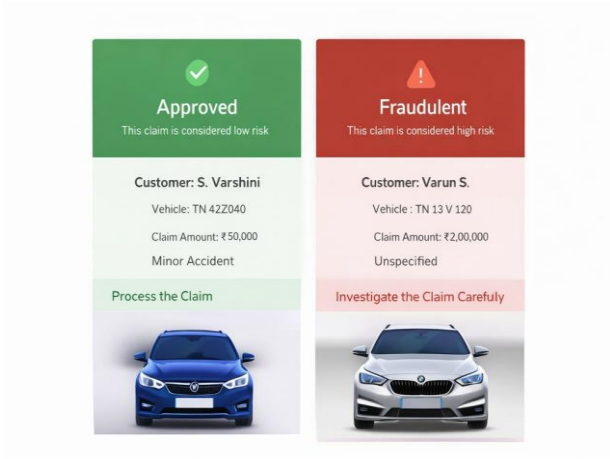

Fig. 7 Fraud Prediction Result Display

V. RESULT AND DISCUSSION

To evaluate how well the proposed car insurance fraud detection system works, the performance of the trained machine learning models was evaluated using four common classification metrics: accuracy, precision, recall, and F1-score. These metrics help in understanding how effectively each model can differentiate between fraudulent and genuine insurance claims. Since fraud cases are usually much fewer than genuine claims, depending only on accuracy is not enough. So, precision, recall, and F1-score were also considered to get a clearer picture of model performance.

Accuracy: Shows the overall correctness of the model. It tells us how often the system makes the right prediction for both fraud and non-fraud cases.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: Indicates how many of the claims predicted as fraudulent are actually fraud. A high precision value means that the model does not wrongly mark many genuine claims as fraud.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall: Measures how well the model identifies real fraud cases. A high recall means the system is able to catch most of the fraudulent claims without missing them.

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score: Combines both precision and recall into a single value. It is especially useful when the dataset is imbalanced, as it balances both false alarms and missed fraud cases.

$$\text{F1-Score} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

The performance comparison of Decision Tree, SVM, Random Forest, and KNN is summarized in Table III.

TABLE II. PERFORMANCE COMPARISON OF MODELS

Algorithm	Precision	Recall	F1-Score	Accuracy
KNN	0.88	0.85	0.86	86.7%
SVM	0.91	0.88	0.89	89.5%
Decision Tree	0.90	0.87	0.88	88.9%
Random Forest	0.95	0.93	0.94	94.3%

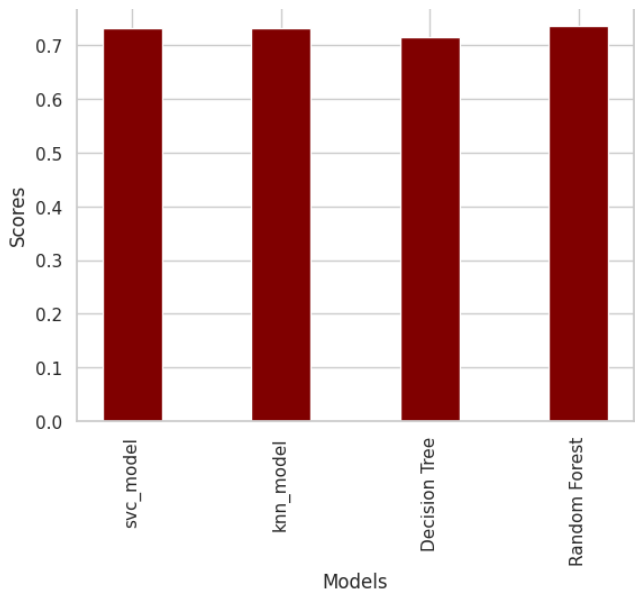

Fig. 8 Performance score comparison of SVM, KNN, Decision Tree, and Random Forest model

Figure 7 compares the performance of different machine learning models used for detecting insurance fraud, including SVM, KNN, Decision Tree, and Random Forest. From the chart, it is clear that all the models performed well, but Random Forest achieved the best overall results.

The comparison shows that machine learning models are capable of identifying fraudulent car insurance claims by learning patterns from previous claim records. Among the algorithms tested, Random Forest stood out because it combines multiple decision trees, which helps in handling complex fraud patterns more effectively.

SVM and Decision Tree also delivered good performance. SVM was able to separate genuine and fraudulent claims even when the data was not linearly separable. Decision Tree, on the other hand, provided simple and understandable rules, which can help insurance officers during the verification process.

KNN achieved acceptable accuracy, but its performance was slightly lower since it relies on similarity calculations and is sensitive to how the data is scaled. Overall, integrating these models into the MERN-based claim system improves fraud detection and makes the claim processing system faster and more reliable.

VI. Conclusion

In this research, a machine learning-based system was developed to detect fraud in car insurance claims using the Kaggle *car_insurance_claim.csv* dataset. The main aim was to check whether a claim is genuine or fraudulent by studying different details such as customer information, policy data, vehicle features, and accident-related factors.

To build the model, different supervised learning algorithms like Decision Tree, SVM, KNN, and Random Forest were trained and compared. The results clearly showed that machine learning can help in identifying suspicious claims more efficiently and reduce the effort required for manual verification.

Among all the models, Random Forest gave the most consistent and accurate results. It handled the data well and reduced the chances of overfitting. SVM and Decision Tree also performed well, while KNN gave reasonable results with proper data preparation.

The final model was connected to a MERN stack web application, allowing users to submit claims and get instant predictions. Overall, the system makes claim processing faster, improves fraud detection accuracy, and provides a practical solution for modern insurance companies.

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